

FFF Printing Fundamentals

Part 3: Dimensional Accuracy

Dimensional Accuracy

- Do not design parts with exact dimensions.
- This is a trap for beginners.
- Why?
 - 3D prints are imperfect.
 - The fit is probably too tight to work as intended.
- Where do these errors come from?
- And how do we fix it?



Dimensional Accuracy

- Sources of error include (but are not limited to):
 - Shrinking and warping.
 - Inconsistent extrusions.
 - Misaligned parts.
 - Bent parts from shipping damage.
 - Incorrect belt tension.
 - Backlash, slop, or loose parts.
 - Moving parts that eventually wear down.
 - Unoptimized slicer settings.
 - Thermal expansion (discussed in Part 3).
 - Vibrations (discussed in Part 3).

Modeling Calibration

- *(Printed size) = (Size) × (Scale factor) + (Constant offset).*
- Simplified model.
 - Look at one dimension at a time.
 - Assume fully functional motion system.
- Scale factor. Error changes with size.
 - Part shrinkage.
- Constant offset. Error is always the same.
 - Extrusion inconsistencies. Over and under-extruding.
 - Surface roughness. Additives like carbon fiber, wet filament.

Scale Factor

- Useful in certain scenarios.
 - Large parts (over 6 inches, 150 mm).
 - Filaments that shrink (like ABS).
- Fixed by adjusting size-related settings in your slicer.
 - XY part scaling.
 - Shrinkage compensation.

Constant Offset

- Very important for any project with assembled pieces!
- **Add clearance** (air gap) between parts that touch.
- Best to add clearance while designing 3D objects.
- Also try adding negative XY offset in slicer.
 - Equally offset all vertical surfaces.
- I consider XY clearance in all of my designs.

Adding XY Clearance

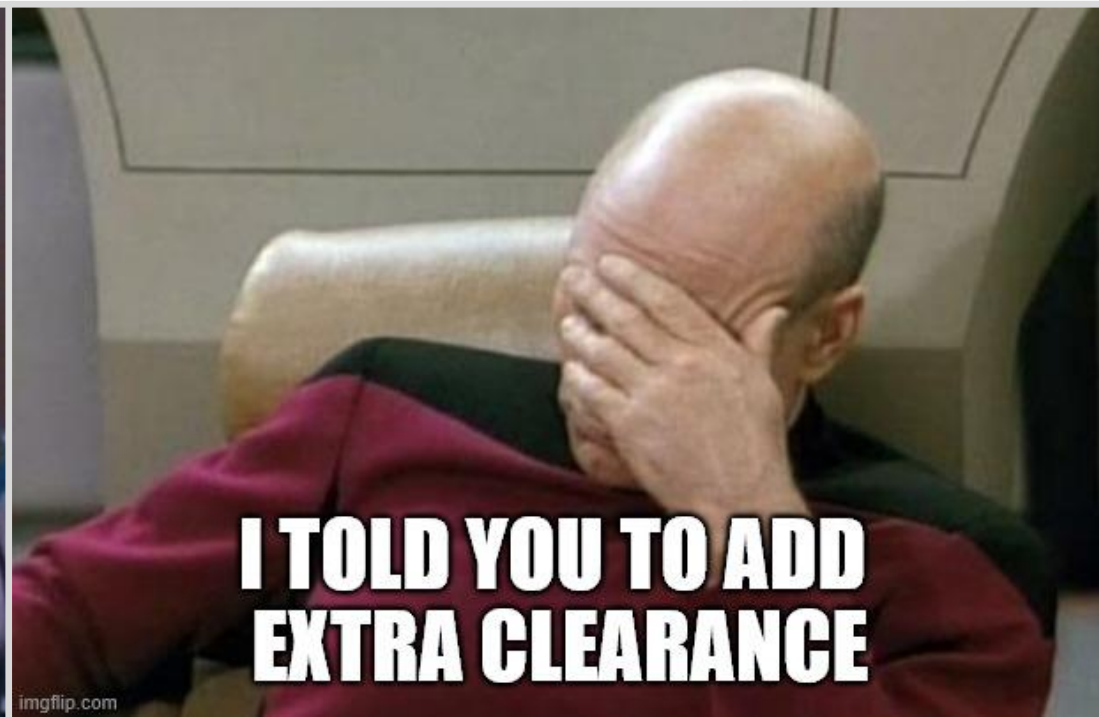
- How much XY clearance?
- It depends, but not very much for modern printers.
- Start with 0.25 mm.
- *Example:* Making a hole for a bolt.
 - Add 0.2 - 0.3 mm of clearance for a close fit.
- *Example:* Making a lid that easily slides into place.
 - Add 0.4 - 0.8 mm of clearance for a loose fit.

Adding Z Clearance

- How much Z clearance?
- It depends ...
- One layer or less for most situations.
- Increase clearance for:
 - Sagging extrusions.
 - Rough bottom surface from supports.
 - High and low spots on print bed.
 - Tilted print bed.

Summary

- If your project has assembled parts ...
 - You sometimes need scale factor.
 - You always need constant offset. Start with 0.25 mm for XY, one layer for Z.



Calibration Prints

- Calibration cubes.
 - 20 mm cube.
 - Labels for X and Y-axis.
 - Tells you constant offset for outside dimensions.
- Clearance test by 3DMakerNoob.
 - Print-in-place clearance/tolerance.
 - Alters the gap between pieces.
 - Feel which pieces move freely.



Calibration Prints

- Califlower by Vector3D.
 - Walkthrough and calculator.
 - Inner dimensions.
 - Outer dimensions.
 - Shrinkage.
 - Skew. (If your rectangles turn into parallelograms).

